B203 Software Engineering: Group Final Project Report

Project Title: Student free Resource material

Module: B203 Software Engineering

https://github.com/Saaqibx77/b203-software-engineering

Introduction

This project is about solving real world problem that most of the students face in the real life when they study in the college. B203 Software Engineering is a good module to learn and come up with the idea of development with the help of our mentors. I have worked on user interface and build some logic that helps to build the report. I have worked and look out for the solution to the real life problem using simple web application with good documentation support. In the real life in college our academic support material is not easily available finding the Previous year questions related to specific module is difficult to find easily and if found they are not well structured or properly available. Some documents are available in weird whatsapp groups and some other social media blogs that supports students but no resource provide with the complete or relevant information.

So I came with the idea of developing a system which is readily available to the students for free access of the material. My idea is to build a centralized easily available resource digital / online web application that is available to all the students. System is created with lightweight accessibility to all with low speed internet connections and low configuration devices. Provided with simple json for data and optimized files for the data storage.

2. Project Planning

To Start with the idea I search out with the online available resources that are providing complete information and the online available of documents with the basic use of Agile Methodology. Used around for build search algorithms that helps to search to data easily.

Content availability not easily available search to documents on various resources is a difficult task.

I had started the building of the idea into reality by giving two weeks break up where I had divided the data collection and front end implementation using basic and simple HTML,CSS

Layout. In the second week I had started implementing the Data Structure (JSON) and the basic javascript for the Logic building. With that data storage and Testing with UI which is easy to use and with that users can upload the contents from the seniors for those people with the use of added technologies.

3. Requirements Elicitation

I found the exact user requirements from new students that are coming new to the college and have discussions related to the problem. Also discuss with my friends who are studying in the same year or few of my seniors about collecting ideas. I had the students to contribute the data, notes, books that can be easily available to juniors. I found various ideas from different class of students and with that I add up my ideas and build basic interface for usage. The System I had required to have dynamic interface where new things, documents can be easily added.

4. Refinement and Specification

Once I had some ideas I thought upon development of technical specifications and creating a single page application which has multiple pages directly connected to the main index or the home page. Multiple pages are built separately books.html, notes.html and a page that provides previous year questions.

Not used SQL because I want my application should run on any platform and easily accessible to all the students and run directly in any system without any setup requirements so easy to use. All data is maintained using JSON Arrays which is very light in weight less memory usages with simple structure of the data.

Data Schema is used which has title of minimum 4 characters , a unique id for the resource.

Coursed drop down where user can select the desired course. Also an option to select the semester that helps to identify the semester requirments, subject which is type string.

5. Further improvments

After developing at the initial I came up with the idea of refinement of my work and to check out the problems with the system and some basic inconsistency in the system.

There must be a control of duplicate entry in the system. Java script arrays for that purpose which is used to easily check the requirements and add on the error message when some duplicate entry found in the system.

Also used the iterative refinement system that helps in the development of the system

6. High-Level Design

The high level design which started by html5 instead of model view controller I had used a simple client layer which user can directly interact with which is elegant and easy to using

simple html and css. Second layer is the logic layer for which I have used the main.js file and get the user input and a search bar which have the logic to search and find the docs easily. The data layer is used which is used to have light weight files that are build using .json file for all document structure.

System starts with books.html which has direct option for browsing the dn static ui also used main.js which uses async fetch() request to the books.json javascript helped in creating dynamic div elements that are inject into the document object model.

7. Low-Level Design

I am not using

any formal backend, our low-level design focused maily on optimizing data and the paring parsing and the database securely within the browser.

For the "Database" I have designed `books.json`, `notes.json` and also the `pyqs.json` as static arrays developed in the javascript objects.

8. Programming Phase: Best Practices

Writing clean and good maintainable code which was a top priority, which is meeting strictly to the software engineering principles discussed in our module.

Our code is strictly divided. HTML handles only the CSS (`styles.css`) handles all visual styling and the animations. Which are build using JavaScript (`main.js`) We avoided inline

The system has the best coding practices of the html, css, js with simple .json file integration.

10. Algorithm Implementation

The most technically complex algorithm in our application is the **Multi-Parameter Search and Filter Algorithm which is very good idea of implentation.**

11. Security Implementation

Care has been taken to maintain the security of the web page with the upload.html is secured by the id and password which is provided to the user when logged in the first time information is stored in the json file for the authentication.

Even without a backend database, security is a paramount concern in modern web applications. The most important point of our application is the `upload.html` form, where malicious users could attempt to inject harmful scripts (Cross-Site Scripting, or XSS) or submit garbage data.

11. Ethical Considerations

As software developers, we must consider the ethical importance of the tools we build. Copyright infringement is a massive ethical issue in out academic career.

12. Testing Phase

No software can be created without the importance of testing, testing is rigoursly done I had implemented at an important level.

1. **Integration Testing:** We tested the linkage between the search bar, the UI dropdowns, and the logic.
2. **Boundary value testing with the Edge Case Testing**

13. Quality Assurance (QA)

Quality Assurance is not includes the bugs but also finding the bugs.

14. Demonstration and Insights Gained

Running the project is incredibly done with all functionality working. As outlined in the lot of insight gained in the develement